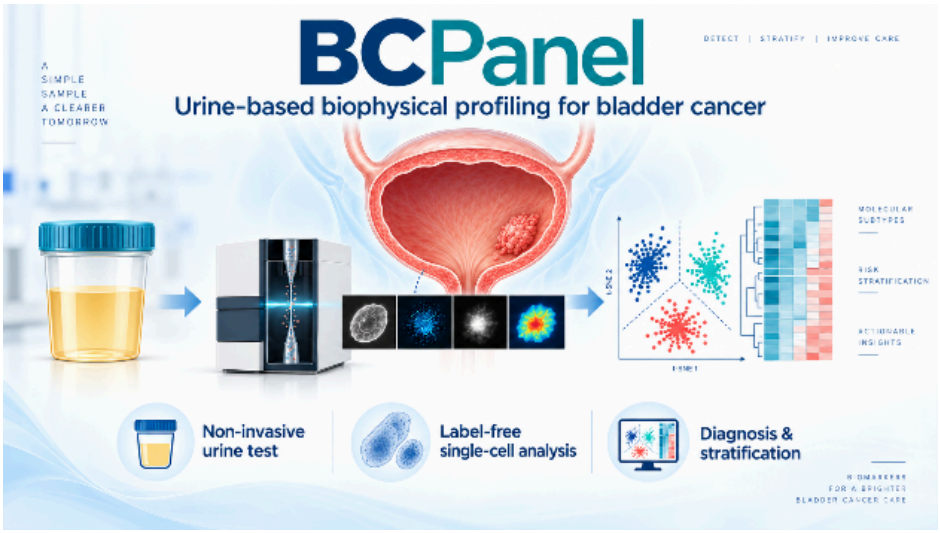
BCPANEL - MULTIPARAMETRIC BIOPHYSICAL PROFILING FOR BLADDER CÂNCER DIAGNOSIS AND STRATIFICATION



Abstract:
BCPanel is a multiparametric biophysical profiling approach designed to support the non-invasive detection and clinical stratification of bladder cancer. The technology quantitatively analyses intrinsic cellular features, including morphological, optical and clustering properties, with complementary biophysical parameters such as cell dry mass and zeta potential.

By integrating these features through multivariate analysis and machine-learning models, BCPanel enables the identification of bladder-cancer-associated cellular phenotypes and supports stratification according to clinically relevant characteristics, including tumour grade, stage, muscle invasiveness and epithelial-mesenchymal transition phenotype. The approach is particularly suited to urine-based analysis and is designed as an objective, quantitative and preferably label-free complement to existing diagnostic workflows

Background:
Current bladder cancer diagnostic and surveillance approaches present important limitations. Cystoscopy is invasive, costly and uncomfortable for patients, while urine cytology has limited sensitivity, particularly for low-grade tumours. There is therefore a need for complementary non-invasive tools capable of providing quantitative information not only on the presence of malignant cells, but also on tumour phenotype and clinically relevant disease characteristics.

Benefits:
Multiparametric, single-cell profiling of intrinsic biophysical features; supports bladder cancer detection and clinical stratification by grade, stage, muscle invasiveness and EMT phenotype; designed for urine-based, preferably label-free analysis; objective and quantitative approach with potential to complement cystoscopy and urine cytology.

Potential comercial use/applications:
Urine-based decision-support tools for bladder cancer detection, stratification and monitoring; complementary diagnostic solutions for urology and pathology workflows; integration into imaging flow cytometry and computational diagnostic platforms; potential adaptation of the biophysical profiling framework to additional oncology applications.

Contacto

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Process number:
PT121776

Patent status:
Patent pending

Cooperation options:
Licensing agreement or research partnership

Technology Readiness Level:
TRL 3 - Experimental proof of concept

Keywords:
Bladder Cancer Diagnostics; Multiparametric Biophysical Profiling; Label-Free Cell Analysis; Cancer Cell Stratification; Single-Cell Analysis; Urine-based Testing; Machine Learning